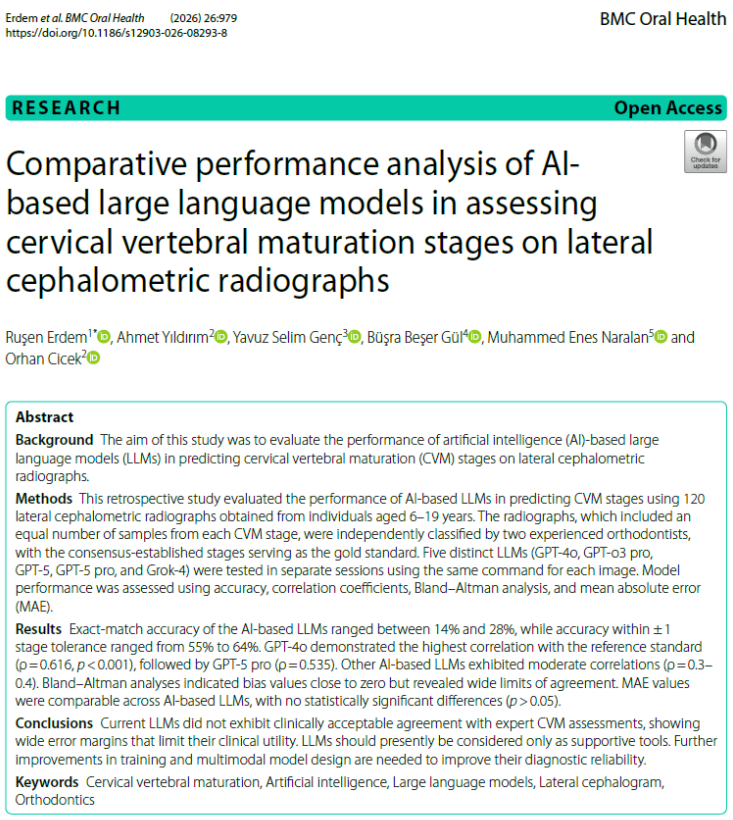
Comparative performance analysis of AI-based large language models in assessing cervical vertebral maturation stages on lateral cephalometric radiographs

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RESEARCH Open Access

Comparative performance analysis of AI-based large language models in assessing cervical vertebral maturation stages on lateral cephalometric radiographs

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Abstract

Background The aim of this study was to evaluate the performance of artificial intelligence (AI)-based large language models (LLMs) in predicting cervical vertebral maturation (CVM) stages on lateral cephalometric radiographs.

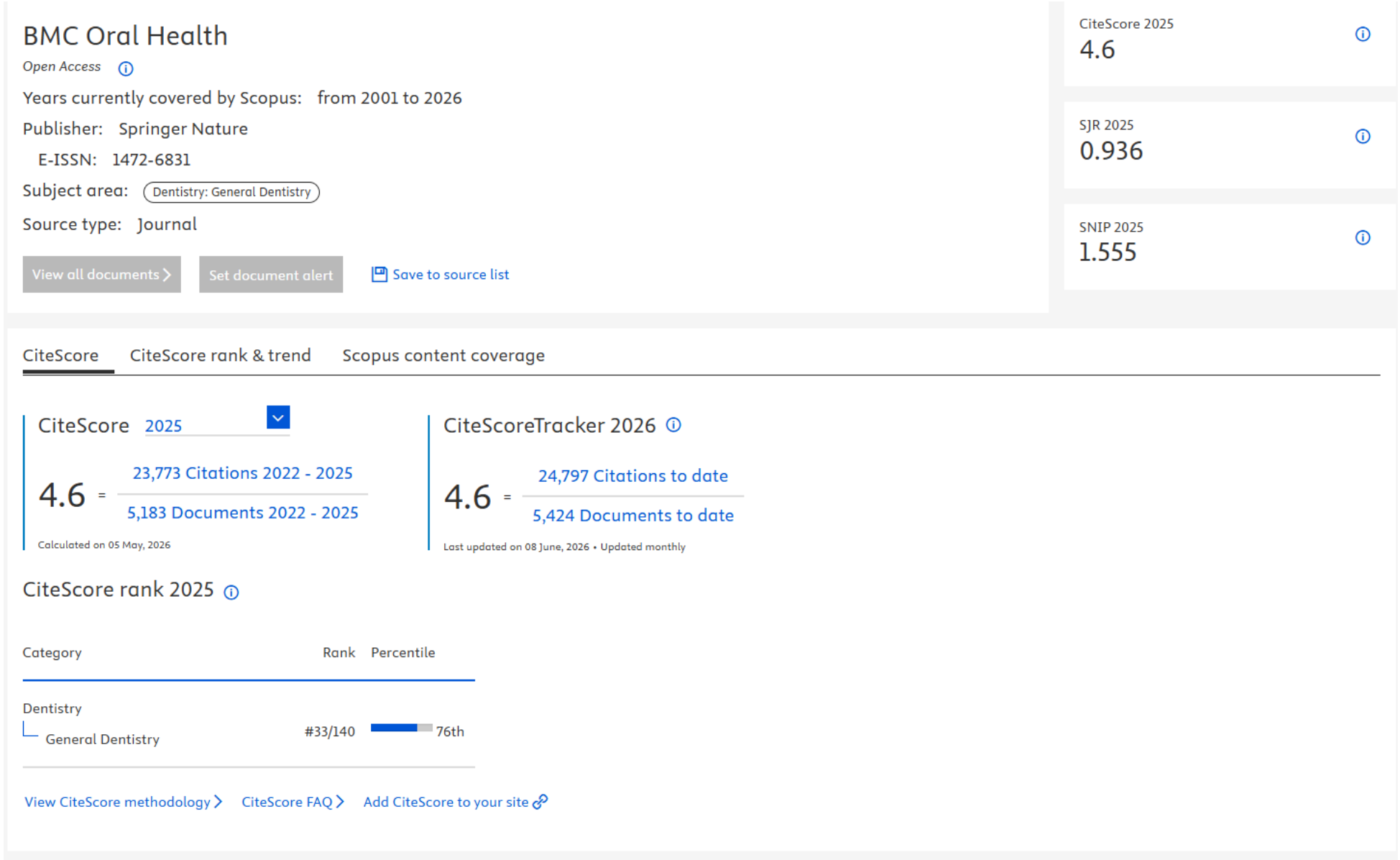
Methods This retrospective study evaluated the performance of AI-based LLMs in predicting CVM stages using 120 lateral cephalometric radiographs obtained from individuals aged 6–19 years. The radiographs, which included an equal number of samples from each CVM stage, were independently classified by two experienced orthodontists, with the consensus-established stages serving as the gold standard. Five distinct LLMs (GPT-4o, GPT-o3 pro, GPT-5, GPT-5 pro, and Grok-4) were tested in separate sessions using the same command for each image. Model performance was assessed using accuracy, correlation coefficients, Bland–Altman analysis, and mean absolute error (MAE).

Results Exact-match accuracy of the AI-based LLMs ranged between 14% and 28%, while accuracy within ±1 stage tolerance ranged from 55% to 64%. GPT-4o demonstrated the highest correlation with the reference standard ($\rho = 0.616$, $p < 0.001$), followed by GPT-5 pro ($\rho = 0.535$). Other AI-based LLMs exhibited moderate correlations ($\rho = 0.3$–0.4). Bland–Altman analyses indicated bias values close to zero but revealed wide limits of agreement. MAE values were comparable across AI-based LLMs, with no statistically significant differences ($p > 0.05$).

Conclusions Current LLMs did not exhibit clinically acceptable agreement with expert CVM assessments, showing wide error margins that limit their clinical utility. LLMs should presently be considered only as supportive tools. Further improvements in training and multimodal model design are needed to improve their diagnostic reliability.

Keywords Cervical vertebral maturation, Artificial intelligence, Large language models, Lateral cephalogram, Orthodontics

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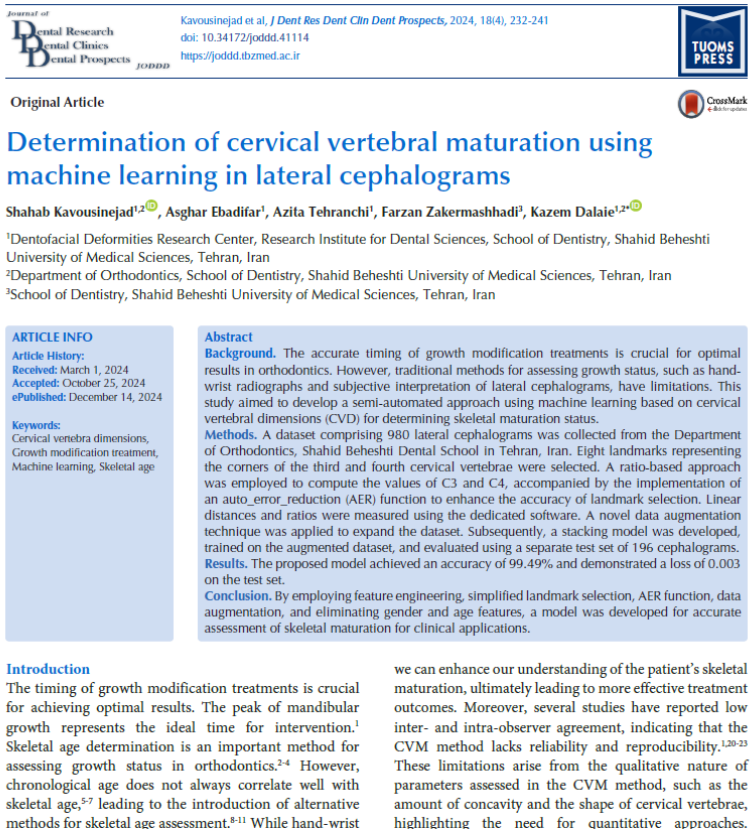
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Original Article

Determination of cervical vertebral maturation using machine learning in lateral cephalograms

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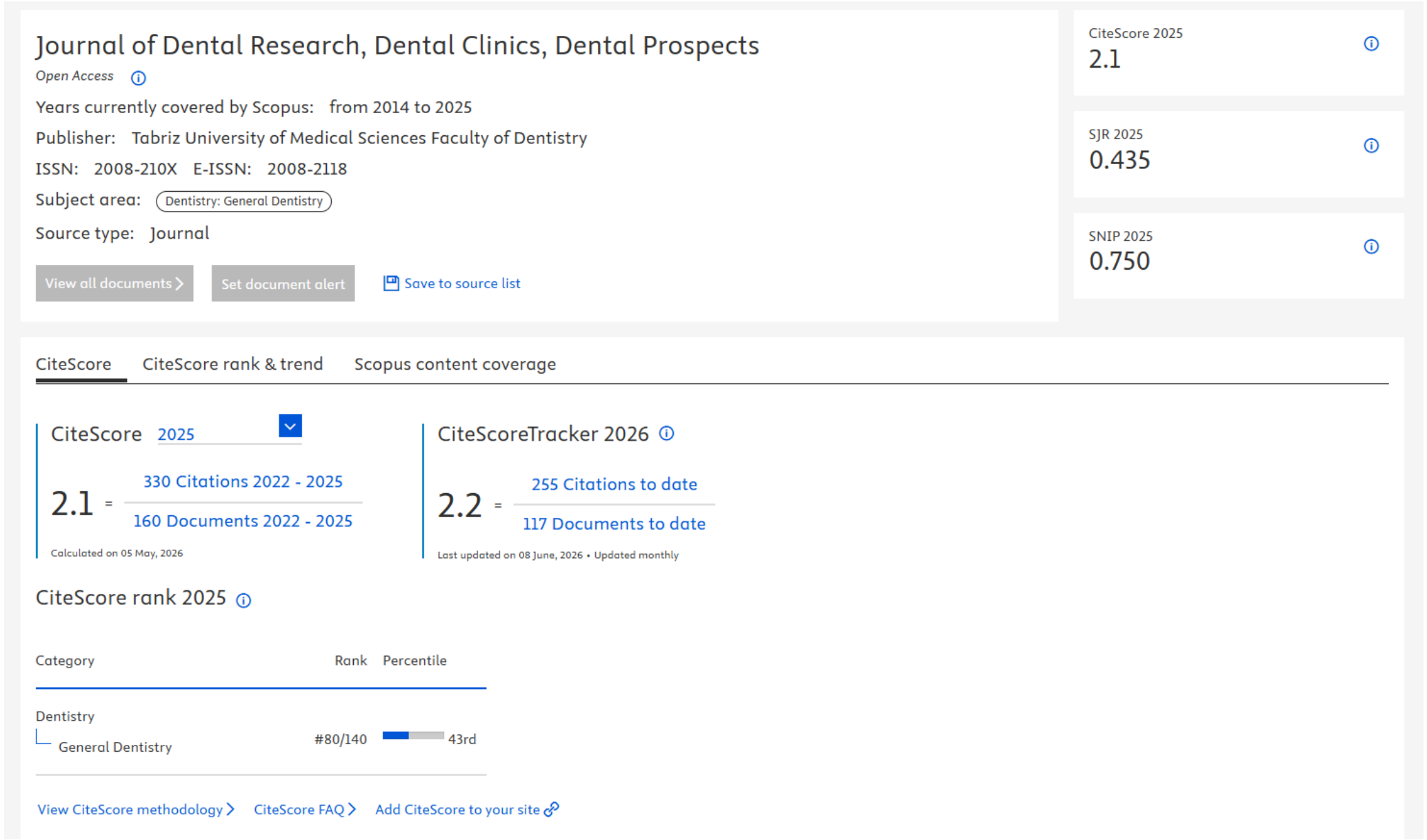
Abstract
Background. The accurate timing of growth modification treatments is crucial for optimal results in orthodontics. However, traditional methods for assessing growth status, such as hand-wrist radiographs and subjective interpretation of lateral cephalograms, have limitations. This study aimed to develop a semi-automated approach using machine learning based on cervical vertebral dimensions (CVD) for determining skeletal maturation status.
Methods. A dataset comprising 980 lateral cephalograms was collected from the Department of Orthodontics, Shahid Beheshti Dental School in Tehran, Iran. Eight landmarks representing the corners of the third and fourth cervical vertebrae were selected. A ratio-based approach was employed to compute the values of C3 and C4, accompanied by the implementation of an auto_error_reduction (AER) function to enhance the accuracy of landmark selection. Linear distances and ratios were measured using the dedicated software. A novel data augmentation technique was applied to expand the dataset. Subsequently, a stacking model was developed, trained on the augmented dataset, and evaluated using a separate test set of 196 cephalograms.
Results. The proposed model achieved an accuracy of 99.49% and demonstrated a loss of 0.003 on the test set.
Conclusion. By employing feature engineering, simplified landmark selection, AER function, data augmentation, and eliminating gender and age features, a model was developed for accurate assessment of skeletal maturation for clinical applications.

Introduction

The timing of growth modification treatments is crucial for achieving optimal results. The peak of mandibular growth represents the ideal time for intervention.[1] Skeletal age determination is an important method for assessing growth status in orthodontics.[2-4] However, chronological age does not always correlate well with skeletal age,[5-7] leading to the introduction of alternative methods for skeletal age assessment.[8-11] While hand-wrist

we can enhance our understanding of the patient's skeletal maturation, ultimately leading to more effective treatment outcomes. Moreover, several studies have reported low inter- and intra-observer agreement, indicating that the CVM method lacks reliability and reproducibility.[1,20-23] These limitations arise from the qualitative nature of parameters assessed in the CVM method, such as the amount of concavity and the shape of cervical vertebrae, highlighting the need for quantitative approaches.

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An Artificial Intelligence Model for the Automatic Classification of the Cervical Vertebral Maturation Stages

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- Journal Name: Dentomaxillofac Radiol
- Scopus Quartile: Q1 Percentile: 97th, 91st, 89th

- https://doi.org/10.1093/dmfr/twag032

Abstract

Objective

To develop an artificial intelligence (AI) model for automatic classification of cervical vertebral maturation (CVM) stages on lateral cephalometric radiographs (LCR).

Materials and Methods

Overall, 1140 LCRs were labeled and classified according to the CVM user guide by McNamara and Franchi using a 2-stage pipeline. In the first stage, the LCRs were cropped to extract the region of interest (ROI) using the YOLOv11s.pt model. In the second stage, the extracted ROI were used to train the ResNet-101 deep learning (DL) model to classify CVM stages. Model performance was evaluated using accuracy, recall, precision, and F1-score. Performance comparison between human and AI was conducted by calculating the quadratic weighted kappa and ICC values.

Results

Intra- and inter-examiner reliabilities were measured using weighted kappa values of 0.992 and 0.953, respectively. The ResNet-101 DL model showed 84% overall accuracy, and the quadratic weighted kappa and ICC values between human and AI readings were 0.93 and 0.95, respectively.

Conclusions

The proposed DL model achieved 84% accuracy in automated CVM stage classification. This AI-based framework enhances objectivity and efficiency of skeletal maturation assessment through standardized and reproducible classification. Accordingly, it reduces the influence of subjective evaluation, particularly where clinical workload is high.

Advances in Knowledge

Automated CVM assessment improves objective evaluation of skeletal growth. To our knowledge, this study is the first to integrate YOLOv11-based localization with ResNet-101 for CVM classification, enabling an automated workflow supported by a curated and balanced dataset and a reproducible framework.

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